BGIN Block#11 Meeting Report

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AI and Blockchain Governance

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BGIN Block#11

AI and Blockchain Governance

Date: Monday, October 21, 2024

Location: Washington DC, United States

Duration: Approx. 90 minutes

Summary

Introduction and Background

- The session was part of BGIN's inaugural AI blockchain governance discussions.
- Joseph Beverly, a member of SingularityNet's strategic initiatives team, introduced a new paper on "Enhancing Fairness in Token Voting Systems."
- SingularityNet, led by Dr. Ben Goertzel, has a history of research in decentralized AI. Joey has experience in blockchain systems, reputation economics, and CBDC consultation.
- He emphasized the importance of integrating AI with blockchain governance, despite the complexity and under-researched nature of the topic.

AI and Blockchain Governance Framework

- Blockchain and AI are both transformative technologies but have different strengths: blockchain for security and decentralization, and AI for scalable, intelligent processing.
- Governance integrates these by introducing programmable fairness through AI for decision-making, dynamic incentives, and potentially DAOs.
- The paper's focus was on AI's potential in governance, particularly regarding transparency and fairness in token voting systems.

Current Research and Challenges

- Research is limited on AI's role in blockchain governance, with challenges in defining blockchain governance itself and understanding how AI could support or even enable it.
- Key topics included decentralized governance, dynamic incentives, and pseudonymous identities, and how AI could help manage reputation, voting, and fairness in DAOs.
- Use cases for AI in DAOs include analyzing user participation to infer reputation, and advisory roles in treasury management.

Integration Levels and Practical Use Cases

- Joey proposed a spectrum from zero integration (no AI in governance) to full integration (AI co-governance).
- Presently, DAOs use AI for basic data retrieval, with minimal integration in decision-making processes. Future aspirations involve AI-enabled advisory tools and reputation systems.
- Some DAOs, like MakerDAO, have started exploring AI in governance, though the practical application remains limited by technological and ethical constraints.

Challenges in AI Co-Governance

- The challenge of accountability: as AI tools gain autonomy, determining who is accountable for AI-driven decisions becomes complex.
- Example: the Truth Terminal AI bot, which recently launched a meme coin with funding from Marc Andreessen, raised questions about accountability and autonomy in decentralized AI agents.
- Joey highlighted the risks of AI making treasury or other governance decisions without sufficient safeguards, potentially leading to unintended financial consequences.

Governance Models and Theoretical Context

- Philosophical references were made to Plato's "philosopher ruler" as a conceptual analogy for AI in governance—an idealized intelligent and fair authority.
- Governance systems like democracy and oligarchy were discussed for their flaws, suggesting AI's potential to address governance limitations, though it also carries risks of centralization and power imbalance.

Long-Term Goals and Practical Constraints

- AI could evolve to handle tasks within DAOs, such as proposing or evaluating governance changes, but currently lacks reasoning and contextual understanding.
- Joey cautioned that AI's practical role in DAOs is still limited and mostly experimental, often constrained to narrow applications like reputation management or advisory roles.
- Complex governance issues, such as ethical concerns and accountability, remain unresolved and need further exploration as AI technology matures.

Closing Remarks and Opportunities for Collaboration

- Joey invited interested individuals to contribute to the ongoing research on AI and blockchain governance, emphasizing collaboration and continuous learning.
- The presentation concluded with a note of caution regarding AI's limitations, suggesting that governance in blockchain requires a careful balance of technology and ethical considerations.

Transcript

Okay, I say we get started now.

All right. Welcome everybody to the inaugural AI blockchain governance session or again, I'm super excited to be here. I've been going to BGIN since Block 7 in Canada. And since then, I've enjoyed every single session. I was able to luckily host a session with Mitchell in Sydney. And again, BGIN's gone through some serious waves over the last few years. So again, super excited to be here.

Today we're talking about a brand new paper out of the FASE Working Group under the tutelage of Chloe and Leon. This document was a bit difficult to determine exactly how we approach artificial intelligence, reflecting that, converging that with blockchain, what scale, level, dimension, equally what topic. And even then, blockchain being such a largely well-researched and very scholarly space, what part of governance do we even choose? So the topic we came to is enhancing fairness in token voting systems.

This paper is still quite new. We have one to maybe 15 different references at this point and documentation documents. One key piece of research we'll share with you later on, which is all about blockchain governance, which was actually the hardest piece to define. It's much easier to define AI than it was to define blockchain governance to make it easier to produce this paper.

So let's get going. Breaking up for today, have a little fun. Not saying that all the sessions weren't fun, but I try to make my sessions a bit more enjoyable than usual.

I'm Joseph Beverly. I'm part of the strategic initiatives team at SingularityNet. SingularityNet is one of the oldest, if not the oldest, decentralized AI protocols in the world. Even with having a token and being on-chain by allowing people to use the token to access other AI APIs through our marketplace, we've actually had research and development and technology over the last 15 to 20 years under Dr. Ben Goertzel, who originally coined the term artificial general intelligence back in the early 2000s.

I've done a lot of work in reputation economics, identity, blockchain systems. I worked with CBDCs in Australia and consulted a range of organizations at scale, from ventures through to governments and the like. Jess couldn't make it today due to a full calendar, but hopefully, I represent the best of her.

We realize that artificial intelligence and blockchain governance is a very under-researched topic. When we look at programmability, smart contracts, and the intersection of AI and blockchain governance, it becomes ever more critical to understand the risks involved, even post-implementation risks. So we started this journey looking at blockchain governance in the light of AI and how AI could enable governance—or if it should enable, where do we put it?

Blockchain has been a hot topic for the last 15 years, with Bitcoin's white paper's 15th anniversary coming up on October 31st. Blockchain is a cool concept, but we don't want to create buzzwords; we want to be effective. AI, even cooler. It's going to change everything, from automating taxes to potentially influencing voting systems, or even as simple as chatting with a replica of a historical figure. Artificial intelligence is a nearly infinitely scalable technology, while blockchain focuses on security, with layers above improving scalability and utility.

Governance also plays a role, involving decision-making, game theory, and the complexities of distributed systems. When we look at blockchain, AI, and governance, we see intelligence security, the intelligent design behind security. The best of blockchain is security at its core, while AI brings intelligent performance, even though it may not be fully intelligent. Blockchain and governance itself involve decentralized governance, with challenges around voting and proposals with pseudonymous identities.

Dynamic incentives are where AI and governance align. AI could provide automation, dynamism, and originality, while governance manages incentives and decision-making. When we combine blockchain, AI, and governance, we think of programmable fairness. This involves the security of blockchain, the intelligence from AI, and governance to manage incentives. This is our framework for considering future papers on AI, blockchain governance, and governance in blockchain.

Our purpose is strong—limited scholarly research exists on AI and governance, particularly in blockchain. We're building on work from experts like Guyan, who has contributed significant research in this field. We're considering the ethical limitations and boundaries of AI in governance. Could DAOs, for instance, benefit from AI? Decentralized autonomous organizations are not fully autonomous when humans are involved. Liquid staking, with examples like LidoDAO, shows promising autonomy.

Defining blockchain governance has been challenging, with work from scholars like Lou et al. highlighting various views. Blockchain governance involves structures ensuring blockchain development and use comply with legal regulations and ethical responsibilities. The dimensions include decision rights, incentives, and both endogenous and exogenous governance, which defines decision-making processes within DAOs or decentralized frameworks. Could AI improve these structures? And how might it be used after the initial setup?

We explore topics like lazy governance, AI in automation and decision-making, token voting, and governance mechanisms. We examine where AI fails, even now, and potential uses, like advising or co-governance, or integrating AI to improve data retrieval in governance. The journey involves understanding how to mitigate risks, control decision-making, and integrate fairness in token voting systems through automation and dynamic incentives.

So once we've implemented it, can we remove it if needed? Should we put mechanisms in place? This is a time where AI and different models might have a monopoly if they enter first. Equally, how do you upgrade AI that's directly involved within a smart contract? Should the contract always be upgradable? And if not, how do we supersede it if something goes wrong?

Going back to the question of how integrated AI should be, after this presentation, we'll go through a few questions if that's okay. This is an ongoing paper, so any raw thoughts, ideas, questions, and statements are welcomed.

We need to discuss a scale ranging from zero integration, where a DAO, for example, represents decentralized governance with no AI involvement, to fully autonomous AI governance. With zero integration, is there an apathy towards AI, or are there anti-AI tools embedded? Does a DAO with zero AI involvement thrive while others experiment with novel technologies? On the other end, imagine a DAO that not only writes proposals but also peer-reviews and approves them, and even drafts upgrades for smart contracts or creates new contracts autonomously. This process can take months or years under traditional structures.

Treasury decisions within a DAO, such as managing liquidity, could also be impacted by deep AI integration. However, this could quickly become problematic if AI makes mistakes. How do we manage risks, and where is the boundary for experimentation?

DAOs and blockchain governance face challenges like plutocracy, apathy, and vocal minority issues. These are relevant to AI when considering the centralization of intelligence. New DAO members might struggle to understand or contribute to governance compared to long-standing members. AI could manage reputation systems or credentials by analyzing data points like GitHub contributions. This would help ensure AI serves the system effectively.

AI can aid civil resistance by identifying malicious addresses through predictive analytics, while also presenting an attack vector if AI can manage private keys. Each topic raises questions about AI's role and neutrality. Should AI enhance, review, observe, or mitigate risks?

Key problem statements include: Can integrating AI enhance fairness and transparency in voting, or could it backfire? What tools exist to improve blockchain governance with AI? Are there use cases for AI in lower layers, such as consensus mechanisms, new blockchain initiations, or subnets? Speaking with several large DAOs, I know they're investing in AI to explore these possibilities. As blockchain governance converges with traditional systems, AI tools will play a role.

Future considerations involve the role of AI as it reaches superintelligence. Will DAOs need to manage liquidity manually? How will smart contracts upgrade, given the rapid pace of AI's evolution? While AI technology is advancing rapidly, blockchain isn't as fast in certain aspects.

General intelligence requires reasoning and wisdom, which are novel concepts, but maybe AI can assist in that progression.

Looking at decentralized AI systems, blockchain could improve AI through reinforced learning with historical data, and AI could enhance blockchain by, for instance, dispute resolution. The question then becomes: how do we implement guardrails, ensuring transparency and limitations?

If anonymity is involved in a DAO, could AI still access and use trained data, or is there an option to remain anonymous? Privacy concerns add complexity, and AI models with reserved consent protocols are already exploring solutions.

That concludes the presentation. I would love to hear questions and go back to any part of the deck if needed. If there are no questions, we'll move on to the panel discussion.

Question1

Q: Thank you for the excellent presentation. Both of you did a great job summarizing your results. I found the slide on integration levels particularly interesting.

These topics are broad and global, and I'm curious about where you intend to stop. If you keep expanding, this could turn into a multi-volume book. Even focusing on voting, it's still a broad decision-making area. Do you have a specific use case in mind to test some ideas?

A: That's a tough question. At this stage, it's mostly about surveys and conversations. There are some examples of AI tools being used in DAOs now, like DAOBASE, but the integration level is minimal. LLMs and rags (retrieval augmented generation) are popular for enhancing data retrieval.

Proposals would be a good initial use case. I've been involved in community funding, and AI could streamline that. For credentials, AI could analyze a person's involvement over time, giving an inferred reputation rather than a single-point assessment. In decentralized finance, AI could evaluate creditworthiness without traditional biases.

Endogenous governance could use AI for creating new blockchain tools. Most people rely on standard contracts like Open Zeppelin, and having a repository of AI-reviewed contracts would help rapid prototyping. Privacy, data, and provenance issues, however, need attention.

While there are examples of DAOs like MakerDAO exploring AI for governance, there's a broader, ongoing inquiry into how DAOs handle continuity over generations. AI could provide scalable advisory, co-governance, or even full integration.

Question 2:

Q: At what point does the AI agent act independently from its creator or the data set it was trained on? Is it forever tied to its origins, or can it evolve into an independent entity governing on Arbitrum, for example? Governance is a human invention, so how does AI separate from its creators in this role?

A: Good question. Think of integration as a continuum rather than a binary point. AI's independence might grow gradually rather than reach a set endpoint.

Because if we go away from, let's say, human-only decision making, which is essentially where we are right now, the end state, which is Apple integration, might be inevitable. But as I mean, the process to get to an AI co-governor is extremely difficult and likely impossible at the moment. Implementing something as a co-governor is probably doomed to fail because of AI's technical deficiencies and limited ability to reason and have wisdom.

However, I can imagine a future where HEI might make it possible to have a co-governor that sees and takes opportunities as it analyzes blockchain and other DAOs. But it would come down to training it based on proposals. Creating an advisory tool for a DAO, like MegaDAO, could involve indexing significant posts, keywords, and timestamps that correspond to market reactions and token price dynamics whenever announcements or drafts were put forward.

The challenge at the start involves collecting data, sorting it, and formatting it in a way that is consumable and trains the AI. Over time, AI might serve as an 'organ' process, where it fulfills specific use cases for a period and eventually expands to serve various functions. For instance, one AI might propose, another might evaluate proposals, and a third might draft code for upgrades. Together, these could form an intelligent system, though efficient communication among them remains uncertain.

A key concern in governance is accountability. Politicians are held accountable for their decisions, so if there's an AI co-governor, who would take accountability if things go wrong? This is compounded by AI's scalability, which could lead to a "Sybil attack" problem. The accountability question is also about who controls or "leashes" the AI—if there is a leash, then accountability can be assigned to the person holding it. Without a leash, accountability becomes ambiguous.

In terms of reputation systems, protocols like Deep Funding are exploring ways to identify who makes proposals, who votes, and who allocates governance tokens. This system might extend to AI applications but remains complex. Accountability often rests with the developer or contract deployer, who can manage or upgrade the AI. Still, this brings us to a "leather chair" debate: we can discuss these issues indefinitely without resolution due to our current technological limitations.

In the future, learning, implementing, proposing, and continuously upgrading will likely define this space, though accountability must eventually be addressed. Conversations around product safety and AI ethics are inevitable. The topic of AI and accountability is highly complex, especially when it's intertwined with governance mechanisms like governance tokens, voting systems, and consensus protocols.

Blockchain governance, especially with DAOs, often encounters centralization of power, political governance issues, voter apathy, and power imbalances. Having an AI advisor that provides historical and proposal data is one approach to address information access, though it's only a small part of the governance puzzle. Complex decision-making around liquidity dynamics and consensus mechanisms may also be handled algorithmically in the future.

Governance shouldn't just be about managing technology. It involves ecosystems, values, and objectives that humans want to achieve. Blockchain governance should also consider ethical responsibilities, fairness, and incentives, which are essential to DAO success. These complexities highlight that a simple definition of governance is often insufficient when discussing blockchain and AI.

Moreover, the task of defining blockchain governance is daunting. Writing about AI and blockchain governance can feel like creating a vast compendium of ideas due to the complexity involved. Keeping definitions simple and relevant is necessary to make meaningful contributions to this space. Discussions around these concepts can become extensive, with every small term and concept sparking potential for deeper exploration.

As someone actively involved in DAO governance, I've noticed that it's far more complex than initially thought. Unlike traditional startups, DAOs must consider legal compliance and risks across multiple jurisdictions from the outset. Incorporating AI into DAOs adds another layer of risk, as regulators generally hold humans accountable for decisions. Adding AI raises questions about control, understanding the technology, and its regulatory implications.

In conversations with DAO clients, I've observed a tension between the potential benefits of AI tools and the significant upfront costs and risks involved. While the industry is known for embracing risk and pioneering, discussions often reflect caution about AI. Governance, with its legal responsibilities, is increasingly demanding. The simplistic days of deploying smart contracts and voting on upgrades are behind us.

Maybe where I'll go to here as a pretty easy slide to stick with. The different DAOs I'm working with right now—because I was working in an AI or decentralized AI company—are only really looking at the technology side. The legal and regulatory side is not very much part of the conversations, to be very clear with you, because I don't really operate in that area; I'm not a lawyer. So I don't really get to see those. However, in conversations with them, the DAOs I'm talking to are not new DAOs. They're extremely scaled. They already have all the processes,

procedures, teams, connections, and lawyers. So it's more so that they are willing to spend any amount of money because generally these DAOs have a significant amount of disposable capital. They just continue to churn through or reserve it. Budgets never really exceed them.

So it's not really a problem for these large DAOs, I'll be honest, on the legal side. What they're looking for, especially, is around reputation systems. That's probably the first piece. How do you manage reputation beyond just token voting? Because token-based DAO governance is pretty fundamental right now. You hold tokens, leverage those tokens for five different use cases, and then you gain rewards because you either staked, gained, or received funding or similar. So they're really keen on how they can solve for who is part of their DAO, identifying those individuals in a way that is most aligned with their values, vision, mission, and objectives.

Reputation is one use case, and the other use case is AI as an advisory tool. That's the second most commonly discussed topic. How do you as a user engage with an AI that is, again, quite one-dimensional? It's going to retrieve information based on your query, and it's going to learn and improve based on actions and participation within the DAO itself. Beyond that, we aren't really looking at very complicated examples where, in my admittedly innocent eyes, I can see a need for significant scrutiny around legal aspects. Even now, ethics are a very large part that they are not considering as much. They understand the idea of AI ethics within governance, but they don't really apply a lot of time to that. They're more focused on using AI as an advisor and as an AI-enabled reputation system.

I'm sure you're going to have a lot of work very soon around this, so just hang on. There are going to be some pretty crazy cases that will come up that none of us see happening. I would look at some kind of protocols like DAObase, for instance, which raised $6 million earlier this year. Again, the main use cases are AI as an advisory tool and AI-enabled reputation systems.

That makes a lot of sense. I think what I'm hearing is that scale is very important, and the more scaled you are, the more you can move along the right axis.

Yeah, as well, looking at some of Mitchell's examples that he brought up around AI co-governance as an idea: we're pretty far from that. We're really far from it, both technologically on the AI side, and in terms of the attestations required even to have that happen on the blockchain. The ecosystem itself needs to reach a certain maturity threshold. We are seeing some use cases of AI being embedded into smart contracts, but generally, that's like JavaScript-enabled, such as WebAssembly in the NEAR protocol or ICP. And there are those building off of subnets or layer threes on top of Optimism, Arbitrum, and Cosmos. But most AI systems I'm seeing right now are just GPT wrappers that are doing something, I want to say unoriginal. They're not doing something that is innovative or novel.

Managing AI upgrades as an actual code itself—how does an open-source community that's decentralized, using tokens to vote and increase proposals, handle that? That would be something

quite novel. We're not really seeing it right now, but I feel we have a good finger on the pulse for it happening at some point in the near future.

Yeah, pumped, absolutely pumped.

Do we have any questions or comments from the audience in the room? Otherwise…

I have an interesting case study that's actually been in the market in the last two weeks or so, over the last month. There's an AI agent bot created by a guy from New Zealand called the Terminal of Truths. What happened recently, and it's actually been a bit of a phenomenon in the last week, is that Marc Andreessen gave this AI bot around $50,000 worth of Bitcoin.

Oh, you mean Truth Terminal?

Yes, the Truth Terminal use case, which also launched a meme coin as a form of funding. Now, the direction that this AI bot went was to actually put itself forward as kind of a cult leader, right? And then build a whole bunch of ecosystems and different personalities around that, which I find to be one of the case studies that we could add to the paper.

Just to sort of catch the room up: someone created an AI agent bot, gave it a Twitter account, and it posts regularly. Marc Andreessen gave the AI bot $50,000. The bot then used those resources to launch its own meme coin on pump.fund, which is now trading at around $200 million. That all happened within the last month.

This AI agent has become almost like a point of centralization for a community. And I think it's quite interesting because you can draw analogies between this AI agent, which almost has a computer level of autonomy and centralization, meaning that there isn't really anyone accountable. Yet, people are getting behind this agent, putting their money into the meme coin and supporting its decisions. So I'd like to get everyone's opinion on that.

Just to jump in, it's on the AI agent side. Instead, it's a researcher called Andy—forgot his last name—who was sent $50,000. He ended up making a token and creating another project called Infinite Backrooms. I spent some time looking into Infinite Backrooms recently. It pits two Claudes against each other and creates a virtual CLI between them so they can discuss the bounds of their consciousness. It's quite interesting, but I wouldn't really call it an AI agent. What it does is automate two endpoints, specifically with Claude 3.5, to discuss cognition. It's important to note that AI right now only gives the performance of a conversation. It does not have an actual conversation with you.

I was recently at an event in San Francisco with Preston Warner Ventures and Harvard on human flourishing. One of the key topics from the Stanford social studies professors was that AI right now can only give a performance; it's nothing more than that. With things like Infinite

Backrooms, it gives you an idea of what it may look like, but it's still performance. It has information and retrieves it, responding in the most probabilistic way.

The point I was trying to make is that it's interesting they've used AI almost as the hierarchical top point of the community, in some way to disintermediate having a human accountable at the top of the decisions that this AI is making. Because if Andy were to launch a meme coin and use all his reputation, and it goes poorly, he could potentially face legal repercussions. However, now the decisions are autonomous.

I think the funding went to a wallet that only the AI controls. The researcher did add capability for private key access, but the spending decisions are made by the agent.

It's hard to say, agents right now are still one-dimensional. Having a bot that trades tokens, manages liquidity, and deploys contracts is novel. But the right amount of hype led to it becoming what it is now. Beyond price dynamics, the setting up of the research around the interaction between two Claudes to discuss cognition is most interesting.

Yes, very interesting use case. I hope everyone checks out Infinite Backrooms—it's super cool.

Maybe to continue, building on a point on the difference between expectations, theory, practice, and the risks involved, if you could go to the slide with the three circles. When I saw this slide, the first thing that came to mind was Plato's *The Republic*, which I happen to be reading now. It talks about justice, and they try to define what justice is. This debate is also about governance representing humans, as Mitch mentioned. Blockchain is just technology, and we add another component, which is AI—a new beast. If we don't know what fairness means, we will struggle to find the right combination of all these components to do something good.

The author says democracy is actually a terrible system of governance, describing democracy, ruling by the few, tyranny, and oligarchy as flawed. Plato's most advantageous system was the philosopher ruler, which, when broken down, resembles a benevolent AGI in the future. We should read about the philosopher ruler for some familiarities with a benevolent AGI.

If humanity hasn't found answers to these questions, can we find good enough answers for these? My reaction to the case study is from a policymaker's perspective: there is always human accountability.

And we add another component, which is AI—a kind of a new beast. So if we don't know what fairness means, then ultimately we will be struggling to find the right combination of all these components to do something good. In Plato's book, even though he advocated for democracy, voting, and fairness, there's a conflict; his own account argues that Socrates, who was a proponent of these ideas, was ultimately sentenced to death by the democratic state of Athens that he was upholding. So, I wonder whether we can encompass all these challenges and if, after

2,000+ years, humanity has not solved or found answers to those questions, whether we can find good enough answers to these.

You'll find, when you review *The Republic,* Plato actually says democracy is a very flawed and terrible system of governance. He describes four systems of governance that fail: democracy (rule by the many), oligarchy (rule by the few), tyranny (absolute control), and democracy again. The fifth and final option Plato put forward, which he believed was the most advantageous or benevolent, is the idea of the philosopher ruler. When you break down Plato's concept of the philosopher ruler, it resembles an AI-like entity that controls information, uses censored knowledge, and focuses on STEM-based education.

He also discusses how one accumulates things on top of the soul, illustrated by a mythical beast (I can't remember the exact name). Although Plato was deeply flawed himself, saying that everyone was born equal while owning slaves, we can't rely solely on classical examples. But I hear you on this, and I'd recommend everyone look up what the philosopher ruler looks like and how it was written about because you'll notice some familiarities with a benevolent AGI in the future.

My reaction to the case study being presented is that from a policymaker's perspective, there's ultimately human accountability. To analogize it with Tornado Cash, for example, a cypherpunk might argue, "They only published code; it's not their control what others did with it." Meanwhile, another perspective might argue, "They should have anticipated potential misuse and implemented KYC or similar measures." Similarly, you can imagine a debate here where someone might say, "Oh, he didn't know the AI was going to create a meme coin or whatever." But then, I could imagine a regulator saying, "He gave it money. He should have narrowed its potential actions."

Ultimately, a regulator's instinct is to find a human accountable. I don't think it will be acceptable to point to an AI or bot and say, "It wasn't me; it was the bot." This is why, given the nature of my work, I always think through scenarios of taking on maximum accountability and responsibility, since by the time a regulator is involved, that's often the expectation.

I do like the experimental nature of it. It's certainly attention-grabbing and gets the imagination going.

In the famous words of my mother, "Be careful what you put on Facebook because it's there forever." Not always true, but on the blockchain, it is. Blockchain is forever. It creates a record; nothing gets erased. So, involving AI in blockchain requires caution. Cases will arise where there will be meme coins, malicious contracts, and malicious AI out there. This will happen in an open-source, public environment with limited guardrails and a significant inclination towards anonymity and pseudonymity.

Unfortunately, these things will happen. The key will be having a seamless, scaled timeline for integrating AI and being extremely pragmatic and proactive about it. Last thing you want is an AI with access to private keys that dumps your project. It'll likely lose people money rather than steal it because of its limited intellectual capacity. But even early, basic malicious cases of AI and blockchain will have these consequences.

Also, if you want to do some classical literature reviews, let me know. I'll have some time; we can talk about the *Odyssey* at some point.

Yes, we can debate, although I'm afraid it may take a few years to conclude. I'm sure there's no conclusion. That's the great thing about leather chairs—they feel great, and you can sit in them forever.

By the way, speaking of AI, was this presentation written by you or ChatGPT, OpenAI?

Great point. Let me clarify. Let's see what here was written by AI. The images here were too cool for AI, so they weren't AI-generated. This slide, not AI. My LinkedIn photo is actually AI-generated to add a bow tie. These slides weren't AI-generated; they were done to an extent, but I had to review them. AI can be a helpful actor in streamlining research, but it's super unintelligent and often needs extensive revision.

This slide I created myself, though I used AI to overlap words, feeling out what was right over time. Programmable fairness was completely original and was the only thing that made sense, so I wrote it down. And these other slides, made by us, are all by other people.

So, yeah, AI here was used very minimally in the deck. AI is not that powerful yet; humans still need to work. Creating research is difficult—AI is great for drawing out insights and helping streamline, but overall, narrow AI remains limited.

I've been experimenting with different LLMs like Llama. If anyone wants to, it's easy to deploy. Go to a cloud provider, set up Docker, get access to Ollama, and use Open Web UI as the front end. With these tools, you can make your own web UI, query different LLMs, and research. When we talk about AI governance, the question arises of how to determine the appropriate model for a task.

If you're creating a chatbot or doing arithmetic, LLMs currently struggle with it. AI managing finances, for example, is very challenging. Huge strides will come once AI can handle arithmetic well or even gain computer vision.

AI is very unintelligent. It performs actions based on training rather than true cognition.

Are there any comments? We're in the final 10 minutes. Any burning questions or comments? Otherwise, we may round out the session. The room seems content.

May I suggest just one final direction? Maybe state the working group in the document.

If anyone wants to contribute, reach out via Telegram, email, or LinkedIn. Message Mitchell, who has my contact details, or Chloe. When it comes to writing these papers, it's about having fun and enjoying the learning process. So if you're keen on learning, enjoying the process, and researching, please reach out. There's room for many discussions and areas in AI.

While discussing broad AI topics is appealing, it's good to start with a narrow focus, which is why we're looking specifically at token voting systems.

Try Gemini, ChatGPT Cloud 3—great examples. Reach out, and that's all for me.

Exactly, LLMs can't do math independently but retrieve answers probabilistically. We can think of AI through lenses of rational thinking, rational acting, human thinking, and human acting. The journey is complex, so enjoy the process and find what you love.

It's late, so I'll wrap up. Reach out to contribute, and let's give Joey a round of applause. It's deafening. Thank you.